



Application of Modelica/TRANSFORM to system modeling of the molten salt reactor experiment

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ARTICLE INFO

Keywords:

Molten salt reactor
MSRE
Modelica
TRANSFORM
System modeling

ABSTRACT

The molten salt reactor (MSR), being a liquid-fueled reactor, exhibits a strong coupling between neutronics and thermal hydraulics originating from the simultaneous utilization of the fuel as the coolant. Analysis of the dynamic behavior of MSRs thus necessitates the use of coupled multiphysics tools. This work leverages the multiphysics capabilities of the Modelica-based, open-source component library Transient Simulation Framework of Reconfigurable Modules (TRANSFORM) and demonstrates its ability to simulate the behavior of an MSR with the focus on modeling the Molten Salt Reactor Experiment (MSRE). The model employs 1D thermal-hydraulic components with a coarse radial and axial discretization of the active core region approximating the design of the MSRE. The power generation within each node is determined by neutron point kinetics which are modified to account for the effects of fuel circulation. Using the developed model, neutronic and thermal-hydraulic behavior under steady-state conditions is characterized and compared to experimental data and alternative computational codes. Some discrepancies are found but most of them are explained by the modeling choices and uncertainties in input data. Performance of the model under transient conditions is then investigated by simulating experiments conducted on the MSRE; the results are again benchmarked against reference data. Overall, good agreement is attained.

1. Introduction

As with all reactor concepts, modeling and simulation are essential to reactor design and the licensing process for final deployment. However, liquid-fueled reactors exhibit unique challenges due to various strongly coupled physics such as delayed neutron precursor (DNP) and fission product (FP) transport, for which traditional system analysis tools must be modified if possible, or new ones must be created. Flexible and modular tools and workflows, based on modern computational paradigms, are desirable to address the prevailing challenges. They also need to be properly verified and validated to ensure accuracy, reliability and suitability for the intended purpose. As the nuclear industry developed further and an increasing number of light water reactors (LWRs) were constructed and put into operation, a substantial amount of operational and test data, along with benchmarking experiments, became accessible. Conversely, the MSRE provides the only source of experimental data of an operational MSR that can be used to validate modeling and simulation tools for liquid fueled reactor concepts.

One tool which aims to close the identified gaps is TRANSFORM,¹ developed at the Oak Ridge National Laboratory (ORNL) (Greenwood, 2017). This is an open-source component library for analyzing complex

dynamic thermal-hydraulic and other multiphysics systems created using the Modelica programming language. TRANSFORM is a repository of pre-built, re-usable and modifiable building blocks that represent physical systems with a focus on nuclear applications allowing for rapid development of large models. This study leverages the capabilities of the TRANSFORM library to perform system analysis on the MSRE.

In this paper we first provide a brief review of the relevant literature below; in Section 2 we then detail the required underlying theoretical framework for system analysis of MSRs. Next, we present the MSRE model, its structure, inputs and associated assumptions in Section 3. We then discuss the system modeling capabilities of the model under steady-state and transient conditions and benchmark against results to experimental data and results of other software tools. Finally, we conclude this paper with a brief summary and proposals for future work.

1.1. Literature review

Since the 1960s great efforts have been made to develop suitable MSR multiphysics coupling models focused on the interaction between

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¹ <https://github.com/ORNL-Modelica/TRANSFORM-Library>.

Glossary

α	Temperature coefficient of reactivity [pcm/°C]
β	Delayed neutron fraction [pcm]
κ	Thermal conductivity [W m ⁻¹ K]
λ_i	Decay constant of precursor group i [s ⁻¹]
Λ	Prompt neutron lifetime [s]
ν	Average number of neutrons emitted from fission [–]
ρ	Reactivity [pcm]
τ	Fuel residence time [s]
ϕ	Neutron flux [cm ⁻² s ⁻¹]
ϕ_{shift}	Phase shift [°]
ω	Angular frequency [rad s ⁻¹]
ASF	Adjoint Shape Function [–]
C_i	Delayed neutron precursor concentration for group i [atoms]
G	Gain in power [–]
m_j	Fuel mass in node j [kg]
P	Power [W]
Q	Volumetric heat generation rate [W m ⁻³]
SF	Shape Function [–]
T	Temperature [°C]
w_f	Mean released energy per fission [MeV]
WIF	Weighting Importance Function [–]
DNP	Delayed Neutron Precursor
FP	Fission Product
MSRE	Molten Salt Reactor Experiment
ORNL	Oak Ridge National Laboratory
TRANSFORM	Transient Simulation Framework of Reconfigurable Models

thermal hydraulics and neutronics. The first model was proposed by ORNL under the name of Murgatroyd (Nestor, 1962). It encompassed coupling of neutron point kinetics with lumped parameter thermal hydraulics; the loss of reactivity due to DNP drift was modeled in a reduced-order fashion only by taking the core and ex-core residence time into account by using a set of delayed differential equations. Furthermore, ORNL demonstrated inherent stability of the MSRE using a similar model, which discretized the core into multiple regions to improve accuracy (Ball and Kerlin, 1965). The resulting OD reduced-order modeling approach was recently revisited at the University of Tennessee (Singh et al., 2018) where they successfully modified their model to simulate operational anomalies affecting reactivity.

Cammi et al. (2012) at Politecnico di Milano (POLIMI) studied the dimensional effects in the modeling of MSR dynamics and were able to show that the adoption of a 1D model for thermal hydraulics is needed to avoid unphysical oscillations of power and correctly predict the reactivity loss due to DNP recirculation. Cammi et al. (2012) also pointed out that the lack of radial discretization of core power and feedback importance is a possible source of inaccuracy. Chengbin et al. (2016) at Idaho National Laboratory (INL) extended the system code RELAP5 to be suitable for liquid fueled reactors using a multi-channel 1D thermal-hydraulics model coupled with flowing point kinetics and benchmarked it successfully against MSRE data. Years before that Dulla et al. (2004) proposed already a refinement of the neutronics treatment by introducing a quasi-static method for neutron kinetics in a multi-group diffusion approximation with a one dimensional slug-flow velocity field, taking into account the spatial neutronic effects by applying a nuclear importance weighting function throughout the reactor.

The Nuclear Research and Consultancy Group (NRG) extended their in-house thermal-hydraulic system code SPECTRA, which was originally developed for LWRs, by a fully-integrated point-kinetics model to account for circulating fuel (Stempniewicz and Roelofs, 2021). Furthermore, SPECTRA was modified to model FP and DNP transport; a model for isotope transformation to calculate concentrations of important isotopes (e.g. ¹³⁵Xe, ¹⁴⁹Sm) is also available (Stempniewicz and Roelofs, 2021).

Pioneering work enhancing the neutronics treatment by replacing point kinetics with 1D two-group neutron diffusion was done at the Forschungszentrum Rossendorf (FZR) using their in-house LWR code DYN3D, which was modified to the 1D version DYN1D-MSR and successfully applied to several typical transients (Křepel et al., 2005). Similarly, the transition from 1D to 2D thermal-hydraulic models represented not only a step towards higher-order modeling approaches that enable capturing more complex phenomena but also necessitated more computing power. Cammi et al. (2011) presented a 2D axis-symmetric cylindrical two-group neutron diffusion model coupled internally with computational fluid dynamics (CFD). This modeling approach was implemented into the unified simulation environment offered by COMSOL Multiphysics. Zanetti et al. (2014) successfully demonstrated the ability to model the MSRE core as several 3D channels in which the balance equations are solved, while the power profile is given by weighted point-kinetics-like equations. Bajpai et al. (2020) revisited the approach of Cammi et al. (2011) and developed it further to account for 3D neutron diffusion coupled with a 3D two-phase flow regime to predict the behavior of gas bubbles in the core and their impact on the reactivity for molten salt fast reactors.

One of the first 3D neutron diffusion models was coupled externally with a multi-channel thermal-hydraulics model for thermal MSRs was developed under the name of DYN3D-MSR at FZR (Křepel et al., 2007); DYN3D-MSR is based on DYN3D which was modified to enable MSR modeling, leveraging the experience with the predecessor code DYN1D-MSR mentioned above (Křepel et al., 2007).

In recent years, modeling MSR behavior with the help of reduced-order models has also generated increasingly more interest. Argonne National Laboratory (ANL) has developed the System Analysis Module (SAM) for non-LWR safety analysis. It represents ex-core components as simple 1D-pipes, while the fuel channels are modeled in a multi-channel approach with inter-channel heat transfer (Hu et al., 2021). ANL also developed the 3D neutron transport code PROTEUS-NODAL based on homogenized geometry models (Yeon Sang et al., 2018). Yang et al. (2022) coupled SAM and PROTEUS-NODAL with the aid of the object-oriented MOOSE framework. As a result, Yang et al. (2022) presented a state-of-the-art modeling tool capable of simulating accurately and reliably steady-state and transient analyses of MSRs.

Recently, Harvill et al. (2022) showed that GOTHIC, a coarse-grid CFD multiphysics software package developed and maintained by Numerical Applications Incorporated is able to accurately simulate both steady-state and various transient scenarios of the MSRE. Power evolution is modeled via a point neutron kinetics model and the tracer capability of GOTHIC allows for simulation of production, transport and decay of DNPs and decay heat precursors throughout the fuel salt loop (Harvill et al., 2022).

Most of the codes above are either newly developed codes, with the purpose of simulating MSR behavior, or modified codes, which have existed already for longer time and have proven their capabilities applied to LWR behavior. We also remarked that in the realm of high-fidelity codes, open-source tools exist based on OpenFOAM (Weller et al., 1998; Fiorina et al., 2015) and the MOOSE framework (Lindsay et al., 2018, 2022). In contrast with the high-fidelity codes, none of the system codes and reduced-order method tools are openly accessible. This paper proposes Modelica/TRANSFORM as an open-source alternative with the aim of demonstrating its value as a tool for system analysis. TRANSFORM² was specifically developed for Dymola,³ which

² <https://github.com/ORNL-Modelica/TRANSFORM-Library>.

³ <https://www.3ds.com/dymola>.

is a modeling and simulation environment for integrated and complex system in various engineering fields. By utilizing this set of tools we can gather additional benchmarking data for existing codes and compare their performances. It is believed that the potential of this tool extends to downstream utilization in design analysis of MSRs in industry and academia alike.

Using an open-source and standardized modeling language such as Modelica can mitigate the overgrowth of proprietary, internally developed tools, foster collaborative efforts, and allow benefiting from developments in related engineering fields. Furthermore, Modelica supports the Functional Mock-up Interface⁴ standard for model exchange and co-simulation. This allows Modelica-based tools to be easily coupled with various other software tools, enabling interaction of models between different modeling and simulation environments.

2. Theory

While both conventional water-cooled nuclear reactors and MSRs share many similarities in terms of physical behavior, significant differences with regard to thermal-hydraulics and neutron kinetics exist and will be discussed briefly in this section.

2.1. Thermal hydraulics

The thermal-hydraulic phenomena of an MSR require a different approach to simulate compared to conventional solid-fueled, water-cooled nuclear reactors. A significant simplification is due to the fuel salt boiling temperature being several hundred °C higher than the operational temperature; thus unintended formation of void in the active core region due to boiling of the fuel does usually not need to be taken into consideration for the system analysis of the MSR behavior. However, it enables modeling the fluid dynamics as a homogeneous, single-phase medium. Caveat is the flow of void throughout the primary circuit, which is relevant for void transients and is not modeled in this study.

Furthermore, in the case of channel-type MSRs, usually a one-dimensional flow approximation seems to be sufficient on the level of system analysis codes since the flow has a preferred direction (Bestion, 2017). Hence, we can choose to solve the system of equations for fluid dynamics in the typical lumped-parameter approach with finite-volume method and staggered grid scheme for the momentum balances for each node along the z-axis of a cylindrical pipe geometry. This is indeed already implemented in TRANSFORM and ready to be used. For a more detailed explanation we refer the reader to Todreas and Kazimi (2015) and instead focus on a peculiarity of MSR heat-transfer modeling here.

In an MSR fission occurs in the molten salt depositing most of its energy in the fuel and only small amounts in the moderator (Robertson, 1965). Therefore, the fuel serves as a coolant of the moderator at steady-state operation in a design such as the MSRE. The phenomenon of internal heat generation has consequences for the resulting heat-transfer characteristics; hence, an altered heat-transfer correlation is crucial to achieve the correct temperature distribution. Fiorina et al. (2014) proposed a corrective factor added to the traditional correlations for the Nusselt number Nu [-] in the form of:

$$Nu_{Q+j_w} = \frac{1}{1 + \frac{QD}{j_w} \phi(Re, Pr)} Nu_{j_w}, \quad (1)$$

where Nu_{Q+j_w} [-] refers to the complete situation with both internal heat generation and wall heat flux, Q [$W m^{-3}$] is the local volumetric heat generation rate, D [m] is the diameter of the channel, j_w [$W m^{-2}$] represents the wall heat flux and Nu_{j_w} [-] represents the Nusselt number for the wall heat flux alone. For laminar flow, Poppendiek and Palmer (1954) derived $\phi(Re, Pr)$ [-] to be constant and equal to 3/44.

Conversely, for turbulent flow it follows with the delineation of the regimes being $Re \geq 2300$ (Fiorina et al., 2014):

$$\phi(Re, Pr) = 1.656 \cdot Pr^{-0.4} Re^{-0.5}, \quad (2)$$

where Pr [-] is the Prandtl number and Re [-] the Reynolds number. This correlation was calculated for a circular channel geometry. The difference to rectangular channels is negligible in the model allowing the correlation to be suitable for our geometry. The correlation was implemented including both regimes with an interpolation using the hyperbolic tangent function. It was applied to the fuel channel component and the graphite heat transfer volume, detailed in Section 3.2.

2.2. Distributed model of modified point kinetics

Circulation of the liquid fuel introduces three effects on the neutronic behavior of the system; changes to the angular fission neutron emission spectra, shift of the differential scattering cross sections, and a diffusive or advective displacement of DNPs. However, given that the velocity and energies associated with fission neutrons are significantly higher compared to those associated with fluid flow, it can be concluded that there would be negligible to no impact on the differential scattering cross-sections and the angular fission neutron emission spectra. On the other hand, the effects of displacing DNPs by flow are substantial (Wooten and Powers, 2018). Additionally, the time it takes for DNPs to make their way through the core and entire primary circuit is often comparable to the half-lives of the longer lived DNP groups. As a result, a considerable fraction of DNPs decays in the ex-core region and emits neutrons in areas that have low or insignificant neutronic importance (Wooten and Powers, 2018). Thus, one can observe the circulating delayed-neutron fraction to be reduced with respect to the stationary fuel effective delayed-neutron fraction.

The following section gives a brief summary of the underlying theory, for a more detailed explanation we refer the reader to Stacey (2018). We begin with the power-form of the point-kinetics equations (Greenwood and Betzler, 2019):

$$\frac{dP(t)}{dt} = \frac{\rho(t) - \beta}{\Lambda} P(t) + \frac{w_f}{\Lambda v} \sum_{i=1}^6 \lambda_i C_i(t), \quad (3)$$

where $P(t)$ [W], is the time-dependent power, ρ [-] refers to the reactivity, ν [-] is the number of neutrons emitted per fission, $C_{i,j}$ [atom] is the importance-weighted total number of DNPs, β_i [-] and λ_i [s^{-1}] are the delayed neutron fraction and decay constant of group i , Λ [s] is the mean neutron generation time and w_f [J] is the usable energy per fission. Next, we divide the core into j discretized nodes and introduce the power shape function SF , that satisfies Eq. (4) to determine the fission power per node (Greenwood and Betzler, 2019). This has been introduced to add considerations for the modeling of DNP drift and the spatial power distribution in the active core region:

$$\sum_j SF_j = 1, \quad (4)$$

$$P_j = P \cdot SF_j. \quad (5)$$

The shape function SF herein is proportional to the one-group neutron flux distribution $\phi(\vec{r})$ integrated over the node in the reactor. For reasons of simplicity we introduce the adiabatic approximation and assume the neutron flux and thus the shape function to be independent of time. In this work, focused on the MSRE specifically, the axial (along the flow direction) and radial (perpendicular to the flow direction) shape functions were determined with the help of the relative fuel fission reaction rate data from Haubenreich (1964), see Figs. 1 and 2, both graphs are normalized so that Eq. (4) is fulfilled and the integral over the flux equals one.

Furthermore, we introduce the adjoint shape function ASF [-] to account for the neutronic importance of the given node. ASF equals the integrated normalized adjoint neutron flux $\phi^*(\vec{r})$. ASF_j determines

⁴ <https://fmi-standard.org>.

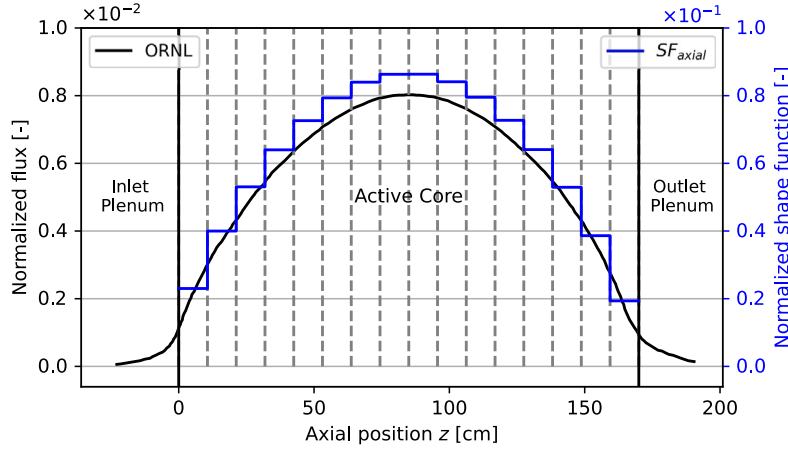


Fig. 1. Axial relative fuel fission density in black (Haubenreich, 1964); axial shape function SF_{axial} for every axial node in blue. Axial position $z = 0$ indicates the bottom of the graphite matrix.

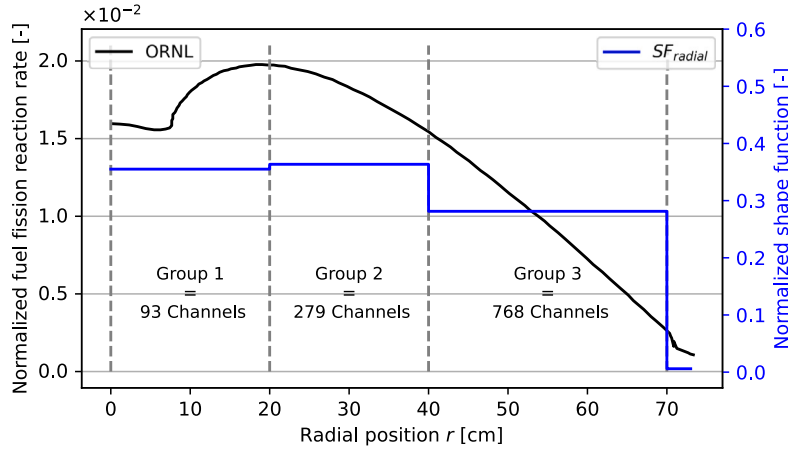


Fig. 2. Relative fuel fission reaction rate in radial direction in black (Haubenreich, 1964); normalized radial shape function SF_{radial} for the three radial groups in blue.

the importance of the neutrons emitted in node j by the decay of the DNP's herein. Implementation of SF , ASF and additional contributions to reactivity denoted as $\Delta\rho$ [-] transform Eq. (3) into the final governing equation:

$$\frac{dP(t)}{dt} = \frac{\rho(t) - \beta + \Delta\rho}{\Lambda} P(t) + \frac{w_f}{\Lambda v} K \sum_j \left(\sum_{i=1}^6 ASF_j \cdot \lambda_i C_{i,j}(t) \right), \quad (6)$$

where the $\Delta\rho$ consists of external reactivity insertions ρ_{ext} and temperature feedbacks. These are determined via reactivity coefficients α [$^{\circ}\text{C}^{-1}$] multiplied with the weighted temperature difference $\Delta\bar{T}$ [$^{\circ}\text{C}$] between the nodes temperature T_j and a given reference temperature T_{ref} :

$$\Delta\rho = \rho_{ext} + \alpha^f \Delta\bar{T}_f + \alpha^g \Delta\bar{T}_g, \quad (7)$$

$$\Delta\bar{T} = \sum_j WIF_j \cdot (T_j - T_{ref}), \quad (8)$$

where the weighting importance function WIF [-] can also be found in ORNL documentation (Engel and Haubenreich, 1962). In a first order approach we approximate ASF to be:

$$ASF_j \propto \frac{WIF_j}{SF_j}, \quad (9)$$

which enables us to deduce ASF from ORNL documentation. Note that, similarly to the shape function, WIF and ASF must satisfy:

$$\begin{aligned} \sum_j WIF_j &= 1, \\ \sum_j ASF_j &= 1, \end{aligned} \quad (10)$$

and finally one must introduce the normalization factor K [-] in Eq. (6), which can be derived from first principles (Akcasuh, 1971) as:

$$K = \frac{1}{\sum_j ASF_j \cdot SF_j}. \quad (11)$$

One way to account for the phenomenon of circulating fuel is to introduce the scalar transport equation for DNP's, in which we neglect the diffusion term and rewrite it for discrete volumes. For one-dimensional approach the balance equation follows as:

$$\frac{dC_{i,j}(t)}{dt} = \frac{v\beta_i}{w_f} P(t) \cdot SF_j - \lambda_i C_{i,j}(t) + \dot{c}_{i,j}^{in}(t) - \dot{c}_{i,j}^{out}(t), \quad (12)$$

where $\dot{c}_{i,j}^{in}(t)$ is the incoming and $\dot{c}_{i,j}^{out}(t)$ [atom s^{-1}] the outgoing precursor atom flow rate of DNP's of group i in node j determined by the fluid mass and momentum balances using an upwind discretization scheme (Ott and Neuhold, 1985).

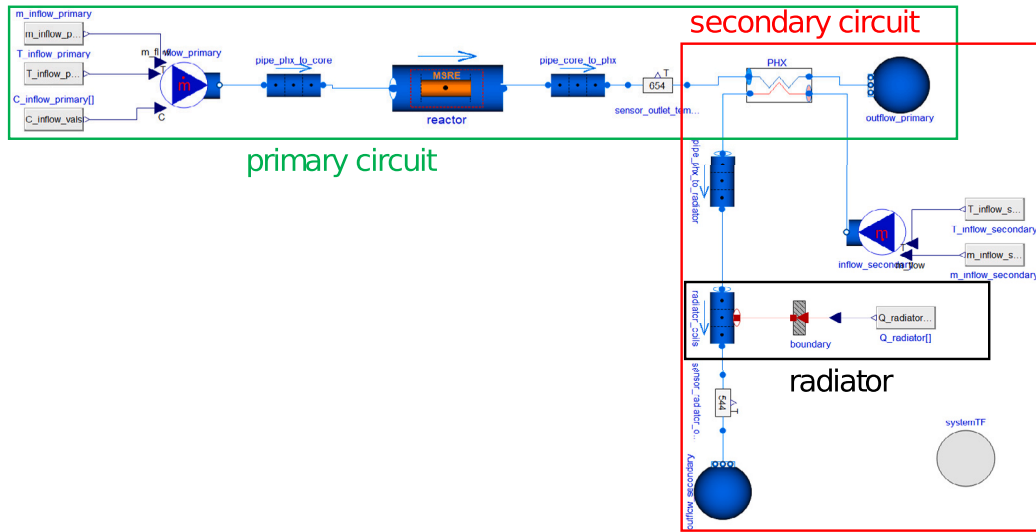


Fig. 3. Overview of the complete Modelica/TRANSFORM MSRE model.

The governing equations (6) and (12) were implemented in the code to determine the spatial power distribution and physical behavior of the reactor; the modified point-kinetics equations coupled to the thermal-hydraulic solver of TRANSFORM thus enable analysis on a system level.

3. The MSRE model

In this section we introduce our MSRE Modelica/TRANSFORM model, providing a detailed description of the model architecture, highlighting its key features and components. Fig. 3 illustrates the component layout of the resulting MSRE model. It can be divided into three circuits (primary, secondary and radiator) indicated in Fig. 3, with the structure of the primary circuit, illustrated in Fig. 4, being similar to the secondary circuit. Note that ex-core components are modeled with less detail, assuming that a finely detailed representation of these components is not necessary for the simulations performed in this work, where the focus lies on the core behavior.

3.1. Primary circuit

In the following the individual components are referred to by a letter indicated in Fig. 4. All fuel guiding components use the thermo-physical properties of the fuel salt $\text{LiF-BeF}_2\text{-ZrF}_4\text{-UF}_4$, which are included in TRANSFORM based on data from Compere (1975). Instead of implementing a pump component, closing the circuit and prescribing the pressure difference Δp at the pump to solve for the mass flow rate, we decided on leaving the circuit open prescribing the mass flow rate. The simulation terminates in the pressure boundary condition (r), in which $p_{BC}^{primary} = 1.5 \text{ bar}$ is set to the inlet pressure of the fuel pump (Robertson, 1965) leaving Δp to be solved by Dymola. This modeling approach also converges to a closed circuit system; however, it fully neglects the pump curve. Hence, the pump in the primary circuit is represented by an ideal flow source (n) for which the following inputs are required:

1. mass flow rate (k), which has a specific value of $\dot{m}_{primary} = 169 \text{ kg s}^{-1}$ under steady state conditions;
2. fuel temperature (l) at the pump under steady-state conditions equal to $T_{primary}^{inlet} = 633^\circ\text{C}$ and for transient analysis it becomes $T_{primary}^{inlet} = T_{PHX, shell}^{outlet}$;
3. concentration of trace substances (m) in units of $[\text{kg}^{-1}]$ at the pump, in this case representing the DNP groups.

The values of (m) are retrieved from the outlet of the primary heat exchanger (PHX), (g) corresponding to the concentration of the six DNP groups ensuring a recirculation of DNPs in the primary circuit. The two 1D adiabatic pipe components (o) and (q) approximate the intermediate piping so that the ex-core residence time for the flowing fuel agrees with the design values. Note that the decay of DNPs in ex-core regions was modeled by implementing a custom sink term S_i [atom s^{-1}] in the governing conservation equations to describe the decay of DNPs of group i in a node j :

$$S_i = -\lambda_i C_{i,j}. \quad (13)$$

The reactor core (p) is described below in more detail.

3.2. Reactor core

The key component of the simulation is the reactor core (p), it consists of 1D pipe components representing the design of the MSRE core, see Fig. 5. The active core region is discretized not only axially but also radially to account for more a detailed representation of the core as discussed below. However, the slim annulus between the graphite matrix and vessel wall was excluded in the final model to prioritize a more thorough analysis of the remaining geometry. The input parameters for the governing equations such as decay constants and effective delayed neutron fractions were obtained from Guymon (1973) and Ball and Kerlin (1965); other important neutronic parameters are listed in Tables 1 and 2. The scaling of SF and WIF within the different core components is summarized in Table 3. The downcomer (p1) is approximated as an adiabatic straight pipe component with 16 axial nodes. The inlet plenum is represented by a 1D adiabatic pipe component (p2); we additionally included a volume with 1 s residence time V_{1s} [m³] to account for the distribution of flow at the lower end of the graphite matrix (Robertson, 1965). The total volume of component (p2) is therefore given by:

$$V_{p2} = V_{IP} + V_{1s} = 0.283 \text{ m}^3 + 0.076 \text{ m}^3 = 0.359 \text{ m}^3. \quad (14)$$

The component is divided into five equally sized nodes. Note that 3% of the total generated power was observed in the lower plenum (Compere, 1975).

The following component (p3) is a 0D mixing volume, which serves as a flow distributor in the model and does not correspond to a physical component of the MSRE. As a result, its volume is negligibly small, and it does not account either for power generation or production/decay of DNPs. The same applies to component (p7), which is designed to

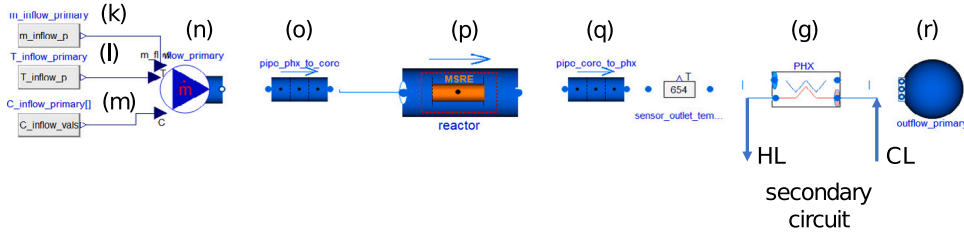


Fig. 4. Model of the primary circuit of the MSRE on the component level.

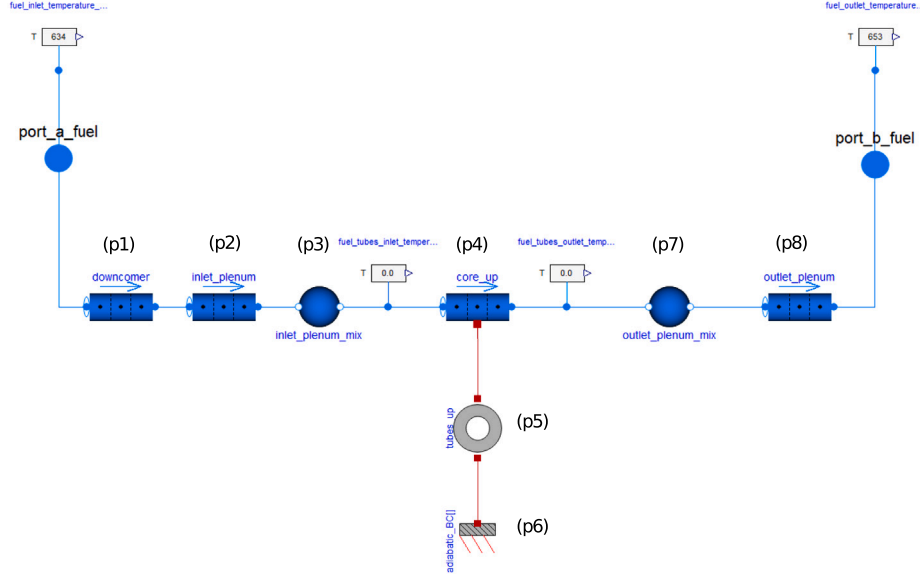


Fig. 5. Reactor component (p) from Fig. 4 on a sub-component level, including the downcomer, inlet plenum, fuel channels and the outlet plenum.

Table 1

Neutronic parameters of the MSRE for the ^{235}U -fueled system (Guymon, 1973; Haubenreich and Engel, 1970).

Parameter	Value
Prompt neutron lifetime Λ [s]	2.4×10^{-4}
Delayed neutron fraction β [pcm]	666.1
Fuel salt temperature coefficient of reactivity α^f [pcm $^\circ\text{C}^{-1}$]	-8.71
Graphite temperature coefficient of reactivity α^g [pcm $^\circ\text{C}^{-1}$]	-6.66

Table 2

Neutronic parameters of the MSRE for the ^{233}U -fueled system (Guymon, 1973; Haubenreich and Engel, 1970).

Parameter	Value
Prompt neutron lifetime Λ [s]	4.0×10^{-4}
Delayed neutron fraction β [pcm]	264.0
Fuel salt temperature coefficient of reactivity α^f [pcm $^\circ\text{C}^{-1}$]	-11.03
Graphite temperature coefficient of reactivity α^g [pcm $^\circ\text{C}^{-1}$]	-5.81

mix the flow from the preceding component (p4), which corresponds to the active core fuel channels. The actual 1120 full, 28 half and 16 fractional fuel channels are represented by 1140 full fuel channels; the geometry of the actual fuel channel was approximated as a rectangular pipe, see Fig. 6. The height h of each rectangular pipe was set equal to the physical one of $h = 1.02$ cm, the width was set to $w = 2.83$ cm and its length along the direction of the flow to 170 cm, so that the flow cross section of 2.89 cm^2 and heat-transfer area of 1309 cm^2 per channel agree with design parameter. It should be noted that the reported fuel channel length varies between ORNL reports; hence, the volume of fuel in the core was used to determine the correct length.

Table 3

Scaling of the input functions for power generation for the sub-components of component (p).

Component		$\sum_j SF$	$\sum_j WIF$
Downcomer		0.03	7.0×10^{-5}
Inlet plenum		0.03	1.6×10^{-4}
Outlet plenum		0.06	7.0×10^{-4}
Fuel channel	Total	0.83	0.999
Graphite annuli	Total	0.05	(1.00 (α^g only))
Total		1.00	1.00
Fuel channel	Group 1	0.32	0.32
Fuel channel	Group 2	0.32	0.32
Fuel channel	Group 3	0.24	0.24

The MSRE core was discretized into three radial groups as a balance between fidelity and computational efficiency. The radial groups consist of 93, 279 and 768 fuel channels, illustrated in Fig. 7; this discretization was chosen so that the total generated power in each group was comparable, see Fig. 2. The friction losses were set so that the flow velocity in the groups agree with measurements of Kedl (1970).

Each fuel channel (p4) is connected to a graphite heat-transfer volume (p5) representing the CGB graphite stringers used in the MSRE (Moore et al., 1964). It was assumed that heat transfer between channels is not needed to be simulated. Hence, the heat transfer volume (p5) is connected to an adiabatic heat-transfer boundary condition (p6).

Finally, the outlet plenum component V_{p8} [m³] agrees with the designed volume plus an additional volume corresponding to a residence time of 1 s, which represents the mixing of flow at the top end of the

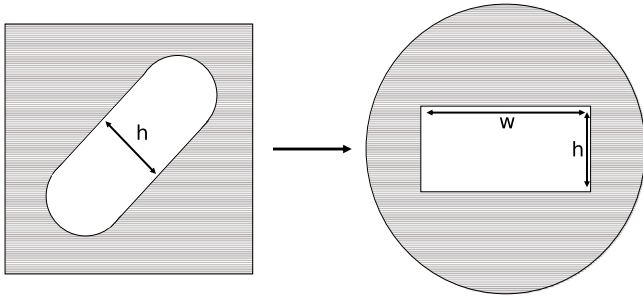


Fig. 6. Geometric approximation for core fuel channels.

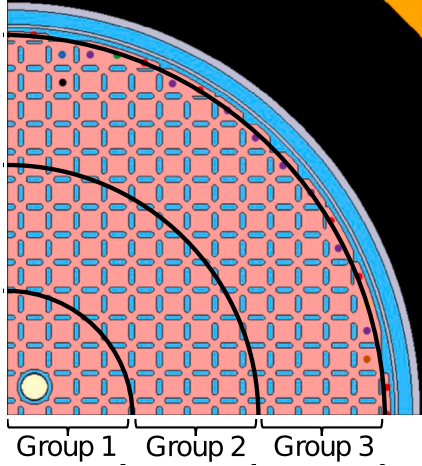


Fig. 7. Top view of a reactor core quadrant at height of the inlet of the downcomer. The radial discretization groups are roughly indicated. Figure adapted with permission from Shen et al. (2021).

graphite stringers (Robertson, 1965):

$$V_{p8} = V_{OP} + V_{1s} = 0.297 \text{ m}^3 + 0.076 \text{ m}^3 = 0.373 \text{ m}^3. \quad (15)$$

The one-dimensional representation of the core allows to model the correct residence time within each component, while the radial discretization allows for a more detailed description of the core.

4. Results and discussion

In this section we present and discuss the results of the studied model. Furthermore, the results are compared with experimental data and results obtained from other codes as well as simple analytical checks.

4.1. Steady-state conditions

The model quickly reaches steady state with prescribed power and cold leg temperatures. This guarantees accurate initial values in subsequent simulations and consequently enables investigation of the model behavior under autonomous transient conditions.

4.1.1. Temperature distribution

The calculated temperature rise in the reactor $\Delta T_{\text{reactor}} = 19.74^\circ\text{C}$ compares favorably with the experimental data of $\Delta T_{\text{reactor}}^{\text{exp}} = 19.6^\circ\text{C}$ (Guymon, 1973). In addition to the comparison with experimental data, we also check against the analytical solution:

$$\Delta T = \frac{P \cdot \tau_{ic}}{\dot{m}_{\text{primary}} \cdot c_p}, \quad (16)$$

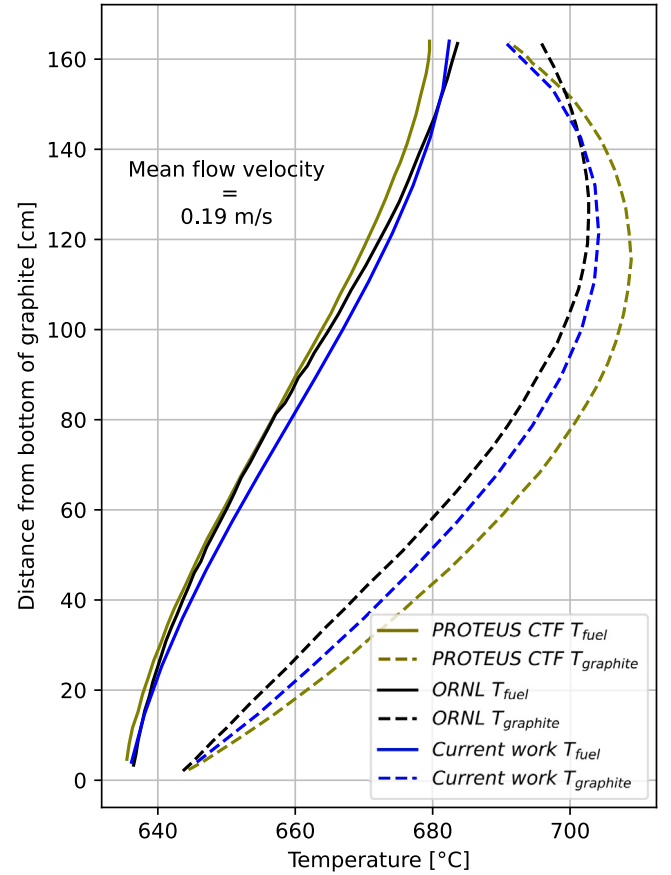


Fig. 8. Comparison of axial fuel and graphite bulk temperature distribution in the hottest channel with ORNL design calculations (Engel and Haubenreich, 1962) and Proteus CTF results (Jaradat, 2021) for power of 8.0 MW.

where c_p [$\text{J kg}^{-1} \text{K}^{-1}$] is the specific heat capacity of the fuel salt. Eq. (16) gives a temperature difference of $\Delta T_{\text{reactor}}^{\text{analytical}} = 19.75^\circ\text{C}$ showing essentially perfect agreement with the result of the model.

Fig. 8 shows the axial bulk temperature distribution of fuel and graphite in the hottest channel of the reactor and compares them to ORNL design calculations (Engel and Haubenreich, 1962) and results from the thermal-hydraulics solver of PROTEUS-NODAL (Jaradat, 2021). Our results were obtained from a model simulating the hottest channel under the same assumptions and approximations as in the overall reactor component. The heat deposition in the fuel salt channel reaches a maximum in the region $r = 18 \pm 2 \text{ cm}$ from the center (see Fig. 7) with a hot channel factor of 2.25. Engel and Haubenreich (1962) report a flow velocity of 0.19 m s^{-1} to calculate the temperature distribution in the hot channel; consequently we observe a laminar flow regime ($Re \approx 900$) rendering the analytical solution of the heat-transfer correlation of Poppendiek and Palmer (1954) to be applicable. We see good agreement with the reference data in the overall trend of the graph, enabling a first successful benchmarking. The remaining discrepancy to the reference data is likely to come from the different graphite heating fractions used in the calculations. Conventional heat-transfer correlations such as Sieder-Tate were not able to predict the correct temperature distribution but tended to show a significantly lower bulk graphite temperature (Çengel and Afshin, 2015). Furthermore, we were able to observe an increasing bulk graphite temperature for neutron irradiated graphite, due to the decrease in thermal conductivity (Nightingale, 1962). Future work for the in-core temperature distribution will include a more detailed comparison of existing heat-transfer correlations for molten salt systems; Cooke and Cox (1973) for example determined a correlation for laminar and turbulent regimes experimentally.

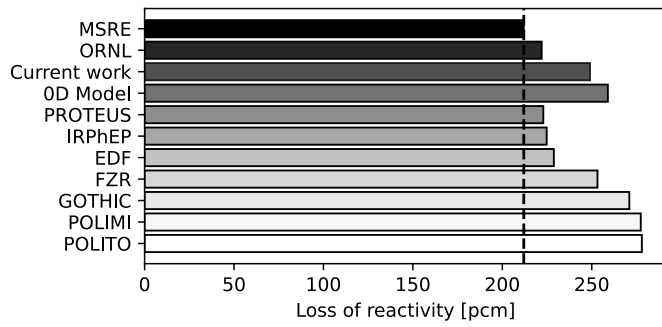


Fig. 9. Comparison of the total loss in reactivity due to fuel circulation in the ^{235}U -fueled system with various other studies of the MSRE (Ball and Kerlin, 1965), ORNL (Haubenreich, 1962), OD Model (Singh et al., 2018), PROTEUS (Jaradat, 2021), IRPhEP (Shen et al., 2021), EDF and FZR (Delpech et al., 2003), GOTHIC (Harvill et al., 2022), POLIMI (Zanetti et al., 2015), POLITO (Spinelli, 2010).

4.1.2. Reactivity loss by fuel circulation

In this work, the reactivity loss due to fuel circulation is calculated by the difference of the effective delayed neutron fractions between static and flowing fuel:

$$\beta^{\text{loss}} = \beta^{\text{static}} - \beta^{\text{circ}}, \quad (17)$$

where β^{static} [-] is taken from Ball and Kerlin (1965) and β^{circ} [-] is obtained from the model. In the MSRE, the total change of reactivity for a ^{235}U -fueled system was measured to be 212 pcm (Ball and Kerlin, 1965). This means that 31.8% of neutrons are emitted in a region of zero neutronic importance and are thus lost from the chain reaction.

Fig. 9 presents the results for the loss of reactivity due to DNP drift of this work for the ^{235}U -fueled system and the results of various other published studies of the MSRE. The present study finds that delayed neutrons equal to 249 pcm are lost due circulation of the fuel. The fractional loss of delayed neutrons by fuel circulation is similar for both ^{235}U and ^{233}U models due to the same mass flow rate. It was observed that the loss of reactivity due to flowing fuel is very sensitive to the flow velocity and shape functions used in the model, especially ASF_{radial} and the weights associated with the plena. Furthermore, the actual distribution of ASF could not be found in the available literature and is therefore subject to a high uncertainty. The approach to model the MSRE in a 0D nonlinear dynamic model was taken from Singh et al. (2018) and implemented in the Dymola modeling environment to serve as a comparison using a substantially simpler model; it calculates the loss of reactivity due to fuel circulation through delayed-differential equations solved under steady-state conditions.

Fig. 9 also shows an overall overestimation of reactivity loss in all presented studies, hence it is speculated that the documented fluid volumes might not be accurate. A larger core volume corresponds to a higher in-core residence time for the DNPs and subsequently a decrease of the fraction of DNPs that decay outside the active core region. At the same time although, measurement techniques in the 1960s might not have been as accurate as of today, an error >10% is considered to be unlikely (Hua et al., 2022).

Another consideration for the inaccuracy is the one-group approximation that is implemented in this work. A Serpent model of the MSRE yielded an overall fast-fission factor $\epsilon = 1.21$, which is expected to be even higher for the plena due to the absence of moderator (Shen et al., 2021). This implies that the channeled region and the plena strongly differ in their neutronic properties. Thus, assumptions underlying the point-kinetics approximation employed in this work, such as globally constant, single-valued neutron prompt generation time, might not be applicable for the MSRE, which exhibits such neutronicly disparate regions.

To examine the results more closely, Fig. 10 shows the spatial distribution of DNP decay in the core compared to design calculations from ORNL (Prince, 1966) and those calculated by the SPECTRA

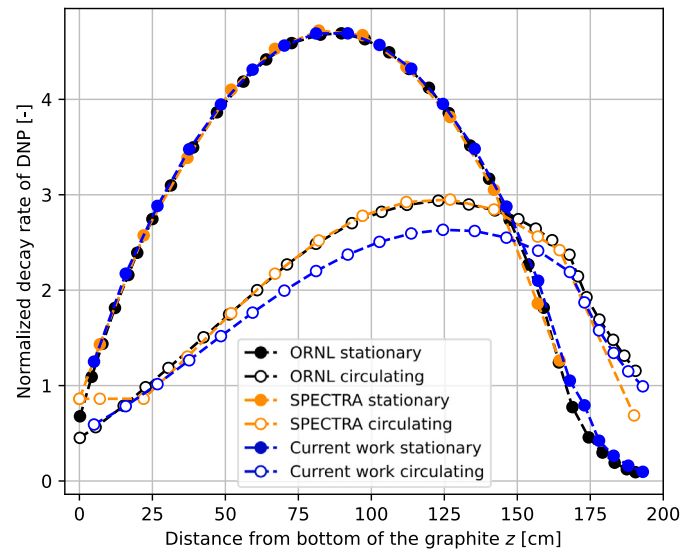


Fig. 10. Decay rate of DNPs along the fuel channel and outlet plenum compared to ORNL (Prince, 1966) and SPECTRA (Stempniewicz and Roelofs, 2021) data for the ^{235}U -fueled system.

code (Stempniewicz and Roelofs, 2021). The values obtained by SPECTRA and this work are normalized so that their respective maximum agrees with ORNL data for stationary fuel. We obtain the values in the following way:

$$\dot{N}_j^{\text{circ}} = \sum_{i=1}^6 \lambda_i \frac{\sum_k C_{i,j,k}}{\sum_k m_{j,k}}, \quad (18)$$

where $\dot{N}_j$ [$\text{s}^{-1} \text{kg}^{-1}$] represents the rate of emitted DNPs per mass m_j [kg] of the fuel of axial node j and summed over k radial nodes. Furthermore, the values for static fuel in Fig. 10 are simply given by the generation rates of DNPs in axial node j . These are proportional to the axial reactor power profile, since all DNPs decay at the location they were born in a static reactor.

For the stationary fuel, the results are similar to the reference data, indicating that the distribution of power and therefore SF are correctly implemented. For the circulating case the above-discussed inaccurate loss of reactivity due to DNP drift is reflected in Fig. 10; the smaller the area underneath the curve for the circulating fuel, the higher the reactivity loss since more DNPs will decay in the ex-core region. Nevertheless, the qualitative course of the graph is similar to ORNL and SPECTRA with the increasing rate of DNP decay along the channel region and the following decrease in the outlet plenum being both captured correctly.

Finally, Fig. 11 presents the axial distribution, including inlet and outlet plena, of the six individual DNP groups for circulating fuel. The distributions are normalized so that the maximum concentration in the channel region of the reactor core equals one; results of PROTEUS-NODAL simulations (Jaradat, 2021) are presented for comparison, these have been digitized and rescaled to account for the homogeneous porous medium approach used therein. Furthermore, results of the a semi-analytical solver are also presented. These results have been obtained by a model developed in analogy to Bureš et al. (2023). This model utilizes modified point-kinetics equations similar to the present work for the loss and re-entry of DNPs. Furthermore, this model assumes that all parameters are only functions of linear coordinate z along the flow of the fuel. The semi-analytical results have been obtained by solving concentrations of DNPs under steady-state conditions for the exact same input parameters (shape functions, geometry and nuclear data) as in the present work. It can thus be considered that for $v(r) = \text{const.}$, the current results converge to results

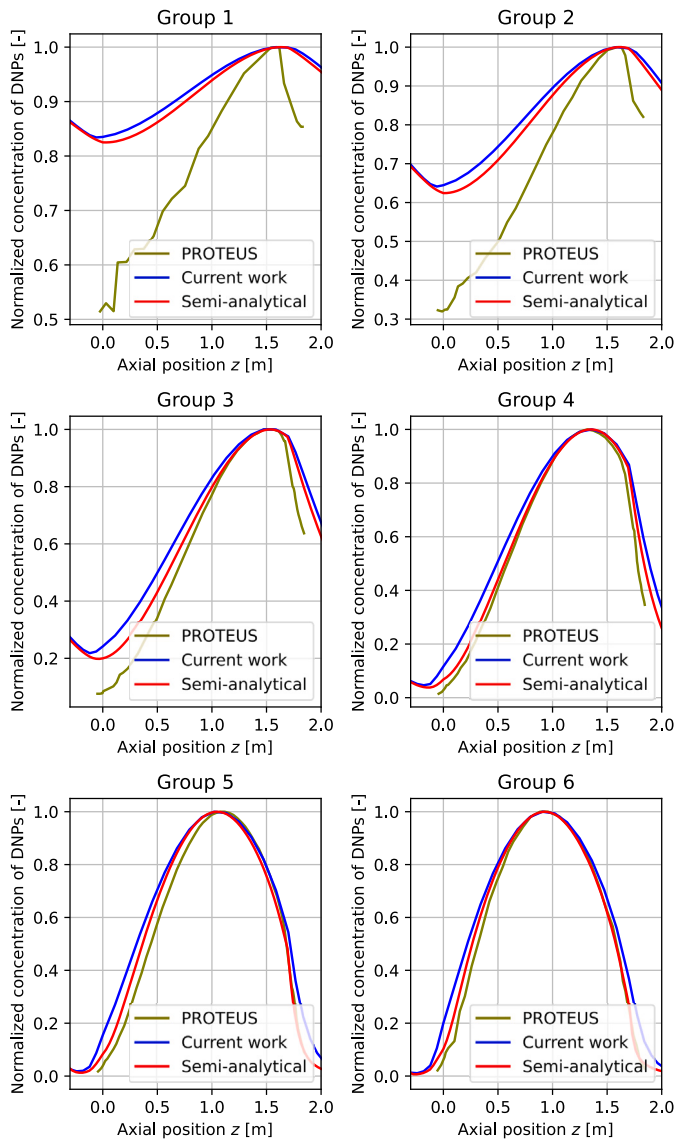


Fig. 11. Normalized steady-state concentrations of DNPs in the reactor compared to PROTEUS (Jaradat, 2021) results and a semi-analytical solver for the ^{235}U -fueled system; $z = 0$ corresponds to the beginning of the channelled region.

of the semi-analytical model for an increasing axial discretization since the semi-analytical model basically represents the analytical solution pointing out the maximum error introduced by the discretization.

The half-lives of DNPs vary between 55.9 s for group 1 and 0.23 s for group 6. The time it takes to make a full loop in the primary circuit is about 25.2 s. Hence, long-lived DNPs are likely to flow one or more times through the complete circuit before they decay and emit a neutron. This is especially visible for group 1 in Fig. 11, for which the difference in concentration between inlet and outlet is small. This difference increases with decreasing half-life, see groups 2, 3 and 4. The concentrations in these groups are skewed towards the outlet due to the fuel flow. The distribution, especially for group 6, is similar to the power profile of the reactor due to its short half-life. The semi-analytic solver does not account for a radial discretization of the core and the accompanying variation in flow velocity; this is considered to contribute further to the difference between our current results and the semi-analytical model. Overall, we see good agreement between the two modeling approaches, therefore we consider the Dymola solution to the equation system for the behavior of DNPs to be verified.

Table 4

Fraction of DNPs which decays in the intermediate piping and PHX ξ_i [%] in the ^{235}U -fueled system for different number of nodes per ex-core pipe component; ex-reactor residence time equals 4.56 s.

Group	1	2	3	4	5	6
2 nodes	4.67	11.05	34.47	67.17	97.81	99.98
4 nodes	5.08	12.00	37.08	70.76	98.66	99.99
8 nodes	5.28	12.48	38.38	72.52	99.01	99.99
12 nodes	5.35	12.64	38.81	73.11	99.12	99.99
Analytical	5.49	12.97	39.80	74.66	99.44	99.99

There are notable differences in the results of groups 1, 2, and 3 between PROTEUS and this study. A comparison between the analytical results for the relative out of core decay:

$$\xi_i = 1 - e^{-\lambda_i \tau_{ec}}, \quad (19)$$

where ξ_i [-] is the fraction of DNPs that decays during the ex-core residence time $\tau_{ec} = 4.56$ s, shows that the displayed PROTEUS results are subject to a significant error which probably originated from digitization of the data. Moreover, based on the analysis of the steady-state profiles, PROTEUS most likely used different nuclear input data than this study. The analytic results and the ones obtained by the current model for each group are compared in Table 4. The results of the model are calculated as follows:

$$\xi_i = 1 - \frac{C_i^{(o)}}{C_i^{(q)}}, \quad (20)$$

where $C_i^{(o)}$ and $C_i^{(q)}$ are the concentrations of DNPs of group i at outlet port of component (o) and inlet port of component (q), respectively (see Fig. 4). Table 4 shows that the difference between the analytical result and the one obtained from the model decreases for an increasing number of nodes per ex-core pipe components. We have deemed the baseline model to be sufficiently converged with 4 nodes per ex-core pipe component.

4.2. Transient conditions

The results of the MSRE model under transient conditions were obtained by a systematic three-step process. First, a simulation is performed until steady-state is reached ensuring a stable system with relevant variables achieving a balanced state; second, the model is initialized with results acquired from the first step to set the stage for the transient simulation. Finally the actual transient is simulated in the third step. This involves dynamically adjusting reactivity or mass flow rate according to the simulated experiment. The implemented reactor kinetics equations together with the reactivity feedback determine the dynamic reactor behavior.

4.2.1. Reactivity step-insertions

We analyze the reactivity insertions for the ^{233}U -fueled system due to the availability and reliability of reference data from ORNL. The reactivity insertions documented in Steffy (1970) are 13.9 and 24.8 pcm for the 1.0 and 8.0 MW power level, respectively. However, available literature (Singh et al., 2018; Stempniewicz et al., 2023) suggests that the values were erroneously documented and the actual values are in a different order, namely 24.8 and 13.9 pcm for the 1.0 and 8.0 MW power level, respectively.

The results of the present work are in good agreement with the experimental data for the 8 MW case (see Fig. 12), although we can observe an overshoot right after the reactivity insertion, similar in height to the SAM model. The detailed view of the dynamic response (see Fig. 13) shows that the sampling frequency of the experimental data is relatively low and is therefore prone to underestimate the initial power peak. The leveling of power is not as strongly pronounced as it is in the SAM model but it appears at the correct power output. The

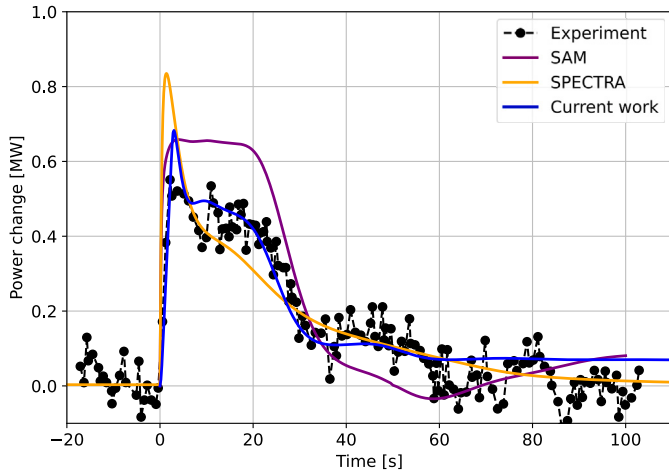


Fig. 12. Change of power after a 13.9 pcm reactivity insertion at an operating power of 8 MW for the ^{233}U -fueled system; comparison of simulated results to experimental data (Steffy, 1970) and results from SPECTRA and SAM (Stempniewicz et al., 2023).

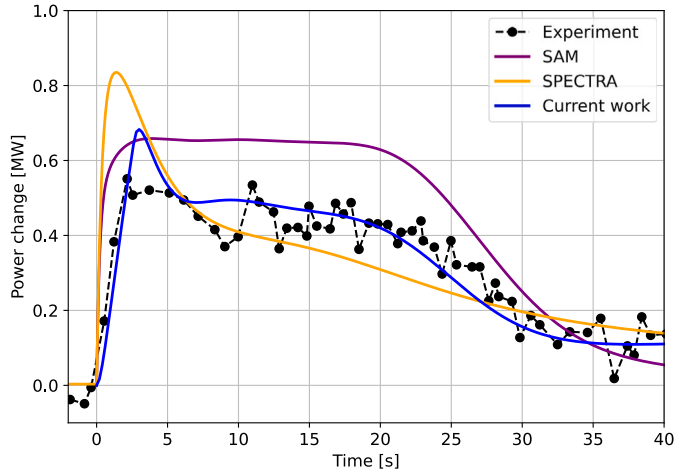


Fig. 13. Detailed view of the first 40 s in Fig. 12.

subsequent oscillation agrees quite well with the experimental data. The offset in power after the reactor return back to steady state in the present work originates from the simplified modeling of the PHX and radiator.

Although the results of the reactivity insertions agree reasonable well with the reference data, the dynamic response of the system to reactivity insertions is not independent of the reactivity loss due to the circulation of fuel. This implies that the ratio of reactivity input ρ_{input} [pcm] with respect to the DNPs decaying in core β_{circ} is increased in the model compared to the experiment:

$$\frac{\rho_{\text{input}}}{\beta_{\text{circ}}^{\text{model}}} > \frac{\rho_{\text{input}}}{\beta_{\text{circ}}^{\text{exp}}} \quad (21)$$

This difference in relative reactivity insertion explains the initial overshoot which would be smaller for a higher β_{circ} . Furthermore, the reference models use an instantaneous reactivity insertion whereas the present model uses a rate of 5.3 pcm s^{-1} (Zanetti et al., 2015).

In the 1 MW case, see Fig. 14, we can observe underestimation of the initial power peak in the present work similar to SPECTRA and SAM results. Furthermore, the experimental data show a negative power change at 200 s after the reactivity insertion. This is neither observed using our model or the SPECTRA one. Fig. 14 shows that none of the models except for SAM correctly predict the oscillations, later than 250 s, which are still visible in the experimental data. Especially the

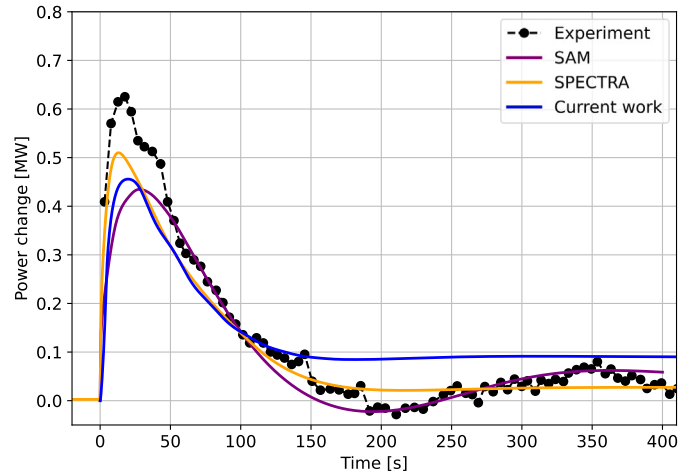


Fig. 14. Change of power after a 24.8 pcm reactivity insertion at an operating power of 1 MW for the ^{233}U -fueled system; comparison of simulated results to experimental data (Steffy, 1970) and results from SPECTRA and SAM (Stempniewicz et al., 2023).

model of this work returns back to a steady state rather quickly (within 150 s); this is due to the modeling approximations for the PHX and radiator.

Overall, our results for the reactivity insertion simulations show a satisfactory agreement with the experimental data from ORNL and the other presented codes. Some code-to-code differences exist but are difficult to explain for a large complex system such as the MSRE. The difference might originate from input parameters, modeling assumptions, nodalization or even different numerical schemes (Stempniewicz et al., 2023). We hypothesize the result of this work could be improved e.g. more accurately capturing the loss of reactivity due to DNP drift and modeling the PHX in a more detailed way.

4.2.2. Flow-rate transients

During operation at zero power (10 W), a variety of tests were performed at ORNL on the ^{235}U -fueled system to experimentally evaluate the inherent nuclear stability of the MSRE. This included flow-rate transients in the primary circuit. The main goal of these tests was to investigate the effects of flow-rate changes on reactivity (Prince et al., 1968).

In our simulation of these experiments, the reactivity coefficients for the fuel and graphite were set to $0.0 \text{ pcm } ^\circ\text{C}^{-1}$, implying the power had no impact on reactivity (zero-power condition). Furthermore, the inlet temperature for the primary circuit was prescribed at a constant level which means the temperature change in the reactor is discarded. The prescribed changes in mass flow rate (see Fig. 15) have been deduced from the pump-speed variations with the help of pump performance curves (Jaradat, 2021). In the experiment, the effects of flow rate changes on reactivity were measured by maintaining a critical reactor during the transient by adjusting the control rod position. The inserted/removed reactivity is then equal to the change in reactivity due to mass flow rate perturbations (Prince et al., 1968). For the model we prescribe the mass flow at the primary pump and hold the reactor critical by reverting the point-kinetics equations (6) assuming steady state in order to calculate the resulting reactivity change:

$$\rho(t) = \beta - \frac{w_f}{P_0 v} K \sum_j \left(\sum_{i=1}^6 A S F_j \cdot \lambda_i C_{i,j}(t) \right) \quad (22)$$

The resulting required response in reactivity to maintain the reactor critical during the start-up transient is shown in Fig. 16. This figure compares our result with reference data; the results are normalized by the magnitude of reactivity loss due to DNP drift at steady state. Thus, the inaccuracies in the steady-state reactivity loss due fuel circulation

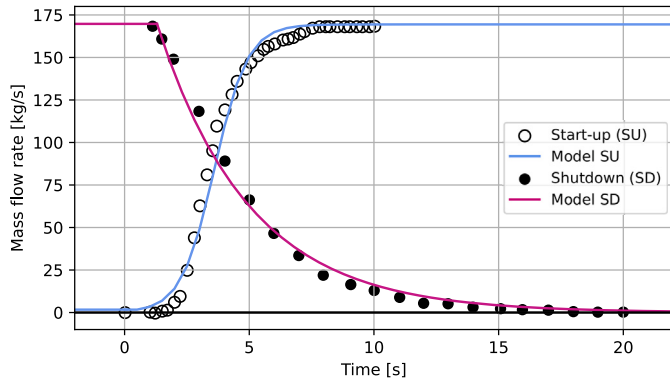


Fig. 15. Mass flow rate variations of the fuel derived from pump performance curves (Yeon Sang et al., 2018) and variations achieved during simulation.

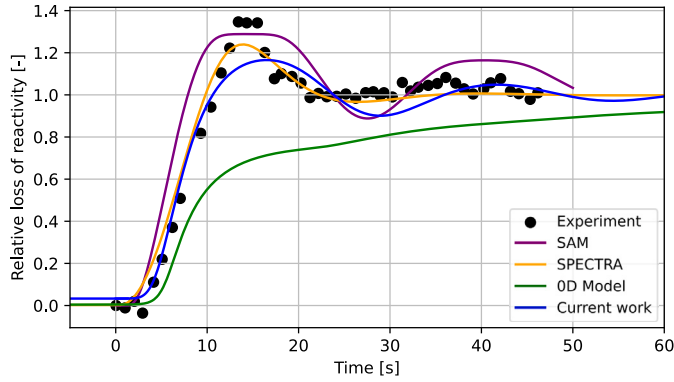


Fig. 16. Calculated normalized reactivity insertion required to maintain a critical reactor during the pump start-up test for the ^{235}U -fueled system, compared to experimental data (Prince et al., 1968) and results from SPECTRA, SAM (Hua et al., 2022) and OD model (Singh et al., 2018).

do not influence the comparison shown. In the experiment the increase in flow rate increases the loss of reactivity and, subsequently, the required external reactivity until the re-entry of DNPs at 13 s. Consequently, the loss in reactivity decreases and stabilizes for $t > 20$ s at the experimental value. In the present model it is not possible to reduce the mass flow rate to 0.0 kg s^{-1} due to numerical issues stemming from the choice of discretization scheme. Thus the initial flow rate for the start-up transient was set to 1%. This implies that the loss of reactivity in the beginning of the start-up transient is non-zero; the loss was calculated to be 4.5 pcm. The reactivity loss due to that initial flow rate is visible at the beginning of the transient in Fig. 16. Apart from this small offset we observe good agreement with the reference data. However, the initial overshoot at 13 s is not as pronounced as it is for the reference data, this is expected to originate from the initial DNP re-entry which affects the height of the peak; we observed an even smaller peak for a residual flow rate of 5%. Furthermore, Fig. 16 also shows the substantially simplified OD model for the flow rate transient which is not able to predict the effects on reactivity due to an increase in flow rate; this might be due to the delayed-differential equation system being not able to capture the time lag of the feedback under transient conditions in the correct way.

The simulation results of the pump shutdown are shown in Fig. 17; the results of the current model exhibit a remaining loss of reactivity due to the residual flow rate at the end of the shutdown transient and do not converge to 0 within the depicted time. The different models show only very small discrepancies with the experimental data for this mass flow rate perturbation. Note that even the OD model is able to capture the effects on reactivity of the pump shutdown.

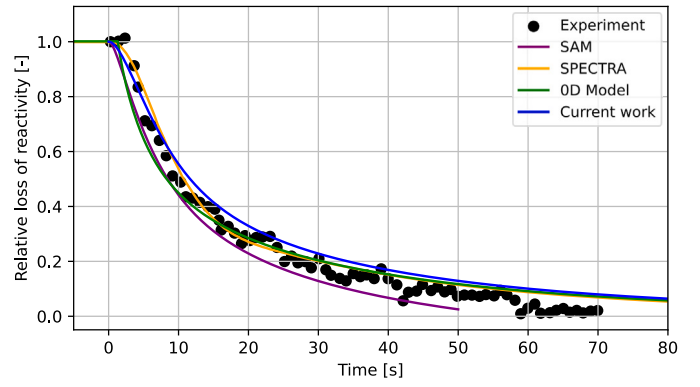


Fig. 17. Decreasing normalized loss of reactivity during pump shutdown test for the ^{235}U -fueled system, compared to experimental data (Prince et al., 1968) and results from SPECTRA, SAM (Hua et al., 2022) and OD model (Singh et al., 2018).

The agreement of both flow rate transients with experimental data is considered to be a further indication that the equation system for transport of DNPs in the fuel is correctly solved in the present work.

4.2.3. Power-to-reactivity frequency response

ORNL conducted experiments by inserting reactivity to the reactor via sinusoidal control rod movement. This type of experiment was conducted for various power levels and both ^{233}U - and ^{235}U -fueled systems. A reactivity insertion amplitude of 10 to 20 pcm was reported for most of the experiments (Kerlin and Ball, 1966). The most satisfactory results were obtained at 5.0 MW and 8.0 MW for the ^{233}U -fueled system. ORNL reported uncertainties for most of the ^{235}U frequency response measurements, rendering it unsuitable for making valid comparisons (Steffy, 1970).

In this type of experiment, the gain $G(\omega)$ [-] at frequency ω [rad s^{-1}] is defined by the change in power ΔP [W] due to the sinusoidal reactivity insertions normalized by the newly acquired mean power P_r [W] around which the oscillations are observed and the amplitude of the reactivity insertion $\Delta\rho$ [pcm]:

$$G(\omega) = \frac{\Delta P(\omega)}{P_r(\omega) \cdot \Delta\rho}. \quad (23)$$

Additionally, the phase shift $\phi_{shift}(\omega)$ [$^\circ$] is defined as the difference between the phase of the reactivity insertion $\phi_{in}(\omega)$ [$^\circ$] and the phase of the response $\phi_{response}(\omega)$ [$^\circ$]:

$$\phi_{shift}(\omega) = \phi_{response}(\omega) - \phi_{in}(\omega). \quad (24)$$

In order to avoid the occurrence of non-linear effects, we have simulated the problem using a sinusoidal reactivity insertion of $\Delta\rho = 1.0$ pcm with varied frequency. The results of the simulations together with data obtained during the experimental campaign are presented in Figs. 18 and 19. We observe good agreement with the experimental data at both power levels. Nevertheless, we see an overall underestimation of the gain $G(\omega)$ at both power levels as well as a deviation from experimental data at higher frequencies. This uncertainty may originate from uncertainties in input parameters of the model. ORNL also reported inaccurate control rod movement and accounted for it by multiplying experimental results by a factor (Steffy, 1970).

On the other hand, we can see a decrease in gain at $\omega \approx 0.25 \text{ rad s}^{-1}$ aligning with a phase shift of $\phi_{shift}(0.25 \text{ rad s}^{-1}) \approx 0^\circ$. This can be attributed to the recirculation of the fuel, which has a natural resonance response ω_r at a frequency determined by the loop transient time:

$$\omega_r = \frac{2\pi}{\tau_{ec} + \tau_{ic}} \approx 0.25 \text{ rad s}^{-1}. \quad (25)$$

The fuel affected by the reactivity insertions re-enters the core one period later in phase with the reactivity insertion ($\phi_{shift}(\omega_r) \approx 0^\circ$) and

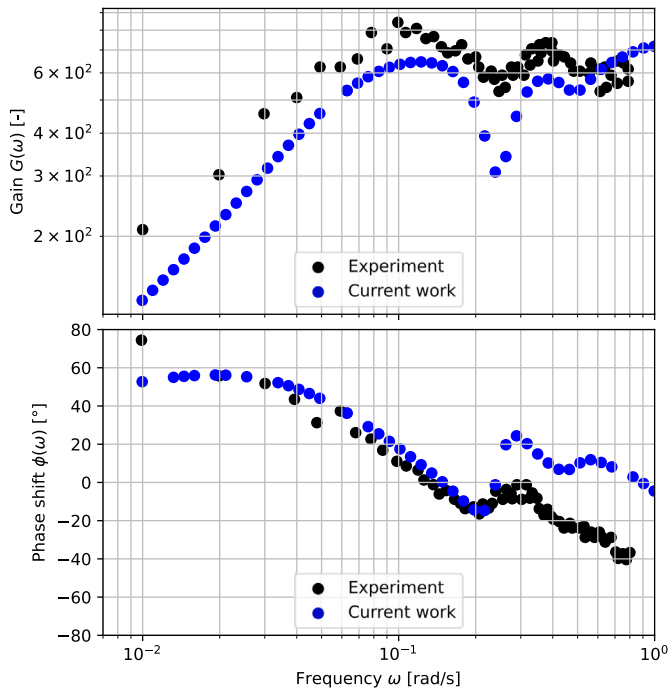


Fig. 18. Response of the 8.0 MW ^{233}U system for a 1.0 pcm sinusoidal reactivity input compared to experimental data from ORNL (Steffy, 1970).

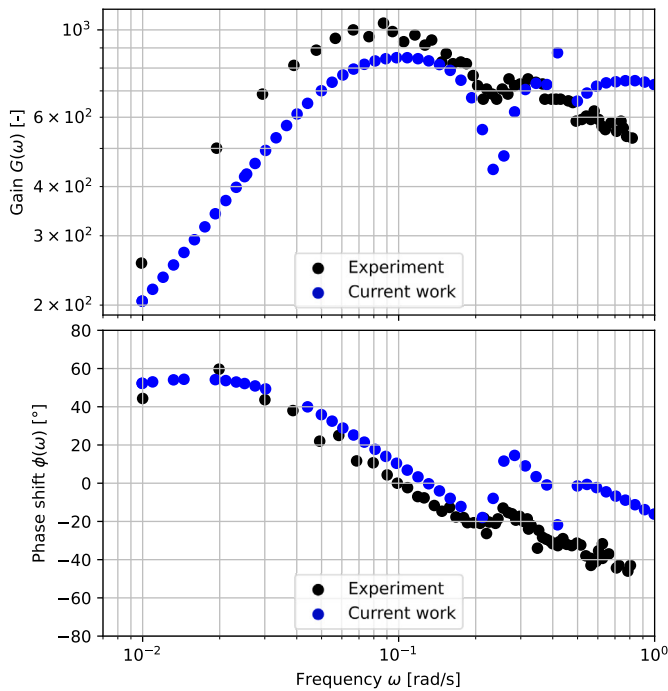


Fig. 19. Response of the 5.0 MW ^{233}U system for a 1.0 pcm sinusoidal reactivity input compared to experimental data from ORNL (Steffy, 1970).

partially cancels the external perturbation, due to the reactivity induced by the negative temperature coefficient of reactivity. The phase shift ϕ_{shift} agrees in both cases reasonable well with the experimental data for $\omega < 0.2 \text{ rad/s}$. The precise origin of the underestimation of gain and deviation in phase shift remains uncertain and will be addressed in future work.

5. Conclusion

The primary objective of this work was to attempt a first effort of verification, validation and benchmarking of the Modelica-based TRANSFORM library for MSR applications. We reasoned Modelica to be a suitable modeling language for the development of a tool for system-level analysis of novel nuclear reactors due to its object-oriented modeling approach enabling rapid and flexible development of computational models.

The modeling technique of this work was mainly developed in-house, leveraging the capabilities of the open-source library TRANSFORM. It encompassed coupling of 1D thermal hydraulics with a distributed model of modified point kinetics, modeling precursor motion through the primary circuit.

Subsequently, the model results under steady-state conditions were investigated, including the system temperature distribution and loss of reactivity due to the circulation of fuel. The change in temperature between inlet and outlet for the reactor operating at 8.0 MW was consistent with the reported data. A comparison of axial temperature distribution in a hot channel model with design calculations and a different computational code showed very good agreement.

The steady-state loss of reactivity due to fuel circulation was found to be higher than the measurement but within the range of comparable advanced system modeling codes. It was observed that the plena have a major impact on the loss of reactivity due to DNP drift due to their relatively large fuel volumes and corresponding residence time; due to the absence of moderator and subsequently the much different neutron spectra therein, the one-group approximation of reactor kinetics might not hold true in these volumes. Furthermore, we observed that even small variations in the input parameters can have significant impacts on the model predictions. The steady-state in-core concentrations of the six DNP groups were similar to the predictions of other codes for the same input parameters. Thus, we consider the Dymola solution of the equation system for DNP behavior to be successfully benchmarked. We were also able to show that the transport of DNPs and their decay in the ex-core components converges to the analytically derived values for an increasing number of nodes per component, verifying our solution method.

Finally, we investigated the models behavior under transient conditions for different power levels, scrutinizing step and sinusoidal reactivity insertions as well as flow-rate transients. We observed good agreement with the reference data for the step reactivity insertions. The flow rate transients also compared favorably with reference data; it was, however, not possible to achieve a flow rate of 0.0 kg s^{-1} (static fuel); this impeded the correct modeling of the flow-rate transients and resulted in a small offset of the calculated reactivity changes. We concluded the dynamic analysis by investigating the reactivity-to-power response, in which we were also able to achieve good agreement with experimental data.

Future work on the present model will involve refinement of radial nodalization, modeling of decay-heat precursors and fission product transport together with their impact on reactivity. Switching from a local machine to a high performance cluster execution might enable the generation of more complex models involving a detailed modeling of the PHX and the overall secondary circuit.

Overall, we found Dymola complemented by the multiphysics system modeling capabilities of TRANSFORM to be a suitable tool for MSR analysis at the system level. We were able to show this using the MSRE as a reference; good agreement was achieved in most situations.

CRediT authorship contribution statement

Lorenz Fischer: Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Lubomír Bureš:** Conceptualization, Methodology, Software, Validation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lorenz Fischer reports a relationship with Seaborg Technologies ApS that includes: employment. Lubomir Bures reports a relationship with Seaborg Technologies ApS that includes: employment.

Data availability

Data will be made available on request.

Acknowledgments

The authors want to acknowledge the support of colleagues at Seaborg Technologies ApS.

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